



New Wingmast Rig for *Wobegon Daze*

(Note: The following is excerpted from an article I wrote about *Wobegon Daze* that was the cover story for the August 2001 issue of *Sailing Central* magazine.—EWS)

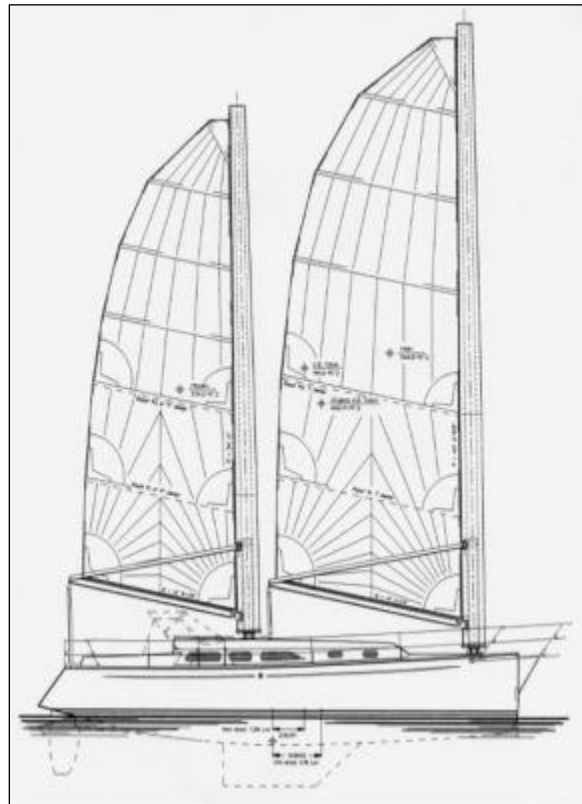


Satisfied with his new hull (see the article *Case Studies in Redesign* on this website for the story of *Wobegon Daze*'s new bow.—EWS), *Wobegon Daze*'s owner, Dr. Blake Cady, next turned to his rig. How could we make it sail better? Although plenty strong, the original carbon fiber masts were a little too bendy, and in gusty conditions they deflected almost too much. Also, because the masts could not turn, sail shape and, therefore, horsepower, particularly off the wind, were not very good. Cady wanted to experiment with a new wingmast design.

My most recent success with wingmasts had been on my open class 60 design, *Project Amazon*, a racing cat-ketch with carbon fiber rotating wingmasts that was an entrant in the 1998 Around Alone single-handed race around the world. Indeed, for *Wobegon Daze*, we used the same builder and some of the same tooling that we had used on *Project Amazon*'s rig. Cady and I spent a full year on design, working out the proper proportions for the sails, masts, and booms, and solving the various problems of control that arise when you let the masts turn.

The advantages for rotating masts are 1) The leading edges of the sails are always at optimum, fair to the wind regardless of the point of sail; 2) Sails can be raised, reefed, or stowed without changing course. This is particularly important when sailing downwind, because you do not have to turn the boat first broadside to the weather, and then into the wind in order to change sail. You hold course and just let the rig weathervane while you do what needs to be done; 3) When sailing downwind, you can set the booms well forward of the beam with the masts pointing aft to get really great lift downwind, which is much more powerful than drag—downwind speed goes way up; and 4) With the booms so far forward and set wing and wing, the boat is naturally stable and extremely resistant to broaching. Uncontrolled gybes almost never happen because forward-set sails cannot get caught by the lee; one sail or the other will naturally pull the boat back onto course should she ever get pushed off by gust or wave.

We finalized the design for the new rig by mid-2000, and construction proceeded through the remainder of the year with the carbon fiber masts being built by Composite Engineering in Concord, MA. The aluminum lower booms and wishbones were built by New England Boat Works in Portsmouth, RI, and the new sails came from Hood Sailmakers in Middletown, RI. Installation of the new rig happened back at Concordia Company Inc., builders of the false bow, on a cold snowy day in December, and that left very little time in the season for any meaningful sailing. We finally got most of the rigging on this spring and could start putting *Wobegon Daze* through her paces.



Each wingmast is built in two parts. The upper carbon fiber wing slips down over a carbon fiber stub mast that is fixed into the boat and extends only about seven to eight

feet above the deck. Two main bearings, custom-built by Harken, Inc. in Pewaukee, WI, connect the two mast parts, allowing the wingmast to rotate. This type of design allows the halyards to exit the stub mast above deck level and remain under tension all the way back to the cockpit without impeding the rotation of the wingmast. Also, with the mast bearings above deck level, there is no open bearing housing that would leak water down below. The deck partners are sealed. Mast rotation is done by control ropes fastened onto quadrants built into the bottoms of the wings and rove through Anderson line-driver winches located in the cockpit.

Another unique feature of *Wobegon Daze's* rig is the combination of conventional booms situated below the foot of the sails, and wishbone booms that connect to the ends of the lower booms and to upper goosenecks on the wingmasts. Inside the lower booms we have Hall Quik Vangs that allow us to control leach tension. Each Quik Vang is attached to a sliding shaft that comes out the boom's front end and is goosenecked onto the wingmast. As you pull or release the Quik Vang line, the boom slides forward and aft about 6" on the shaft, and this causes the aft end of the boom to rise or fall nearly three times that amount, about 18". This magnified vertical movement at the end of the boom is what controls the twist of the sail. Finally, lazyjack lines rigged between the wishbones and lower booms form very nice cradles to catch and stow the sail. At the end of about \$100,000 for the entire rig, *Wobegon Daze* is now the latest incarnation of truly advanced, powerful, and safe sailing. Cady and I will continue to shake down the rig throughout the summer to learn more about its actual performance. I have also started on a new 45' cat ketch sailing yacht design for Cady that will incorporate everything we learn from *Wobegon Daze*.



SAIL Magazine Awarded Sponberg Yacht Design Honorable Mention in the 2002 Freeman Pittman Innovation Award.



SYDI's free-standing rig for *Wobegon Daze* was recognized by *SAIL* magazine for its innovation in rig design in its annual Freeman K. Pittman Innovation Awards. Citing their comments in the April, 2002 issue announcing the winners, the magazine editors said of *Wobegon Daze*'s rig:

"This one hit a soft spot, for, as one judge put it, this was exactly the sort of innovation most apt to engage Freeman Pittman's imagination. The judge's found designer Eric Sponberg's new free-standing carbon-fiber wing mast to be a very well thought out and carefully engineered example of a fast, easily handled rig."

Freeman Pittman was the Senior Technical Editor at *SAIL* magazine for fifteen years. He succumbed to Lou Gehrig's disease (ALS—amyotrophic lateral sclerosis) at the age of 41 in February 1996. I met Freeman at the 1983 Chesapeake Sailing Yacht Symposium where I presented my first technical paper, "Engineering Aspects of Free-standing Masts and Wingmasts." I wrote a layman's version of that paper for the October 1983 issue of *SAIL*, which Freeman edited, entitled "Engineered to Stand Alone." This began a long editor-author relationship that Freeman and I had until his untimely passing.